

Module 1

Language of Mathematics

Aim:

- We will learn about the language of mathematics!
- We will learn how to express our ideas formally and precisely and concisely.
- We will learn new terminology and notation, with the aim of thinking and writing in a mathematical and formal way.
- We will learn about the techniques of writing mathematical proofs
- We will learn the method of proving by Induction

Intro to Mathematical Statements

Mathematical Statements vs Variable Propositions:

Let's begin with understanding notations!

Can we read this statement out loud?

$$\forall x, y \in \mathbb{R}, 0 \leq x^2 + y^2$$

Let's try to break it down in smaller pieces:

$\forall$: Mathematical notation for “For all”

$\in$: Mathematical notation for “Belongs to”

$\leq$: Mathematical notation for “less than or equal to”

$\mathbb{R}$: Set containing all real numbers

Can we read it out loud now?

$$\forall x, y \in \mathbb{R}, 0 \leq x^2 + y^2$$

For all x, y belonging to the set of Real Numbers, the inequality “ $0 \leq x^2 + y^2$ ” is true.

*** See how the “is True” part is silent!*

Let's look at another example:



$$A = \{1, 2, 3, 4\}$$

$$B = \{3, 4, 5, 6\}$$

$\forall$ sets P, Q, U , if $P \cap Q \subseteq U$ then $P \subseteq U$ or $Q \subseteq U$.

$$A \cup B = \{1, 2, 3, 4, 5, 6\}$$

Here's a quick breakdown of the notations used:

$$\text{Null set: } \{ \} \rightarrow \phi$$

$\cap$: This is the notation for intersection of sets. Intersection of two sets results in a set that contains all the common elements between the two sets.

$\subseteq$: This notation can be read as “subset/equal to”. A set A is said to be a subset /equal to a set B if all the elements of a are contained in B.

Can we read the statement out loud now?

$\forall$ sets P, Q, U , if $P \cap Q \subseteq U$ then $P \subseteq U$ or $Q \subseteq U$.

Sets? ✓

Intersects? ✓

belongs to? ✓

subset /equal to? ✓

* Union? ✓

* Null set? ✓

****What are sets?**

Sets are collection of objects without duplicates.

Can we say that the above statement is true?

Remember, the above statement will be considered true if it *always* holds True!

Let's take an example:

Let P be $\{1, 2, 3\}$

Let Q be $\{3, 4, 5\}$

Let U be $\{3, 6, 7\}$

Then $P \cap Q$ is $\{3\}$ and $P \cap Q \subseteq U$

But, $P \not\subseteq U$ and $Q \not\subseteq U$

$\not\subseteq$: Notation for “not a subset”

Hence, we have an example where the above statement doesn't hold true.

We can, thus, say that the above statement is false.

Here's another one:

$\exists$ sets P, Q, U . if $P \cap Q \subseteq U$ then $P \subseteq U$ or $Q \subseteq U$

We used a new notation here:

$\exists$: Notation for “there exists”

The symbol “.” Here represents the phrase “such that”

A “.” after an existential quantifier is read as “such that”

Can we read the above statement in English?

Can we say that the above statement is true?

Remember, for the above statement to be true, we just need one example for which it is True.

Let P be $\{1, 2, 3\}$

Let Q be $\{3, 4, 5\}$

Let U be $\{1, 2, 3, 6, 7\}$

Then $P \cap Q$ is $\{3\}$ and $P \cap Q \subseteq U$ and $P \subseteq U$.

Therefore, we can say that the above statement is True.

Let's look at another example:

$$x^3 - 4x + 2 \geq 0$$

What do we think about the truth value of the above claim? Is it true or is it false?

The truth value of this claim is uncertain and depends on the value of x . If x is 10, then it's true but if x is 1, then it's False.

This leads us to our first definition of the chapter:

Mathematical Statement: *A mathematical statement or proposition or logical statement can be defined as grammatically correct sentence/sentences containing words, symbols and mathematical notations that has exactly one truth value which is either true or false.*

Variable proposition: *A variable proposition is a sentence/sentences the truth value of which depends on the value of a variable inside the sentence.*

We can often represent a claim/proposition as a mathematical statement by using upper case letters along with the involved variable in brackets.

For example:

Let $P(x)$ be the proposition that " $x^2 > 0$ "

Here $P(x)$ is a variable proposition since the truth value of $P(x)$ depends on x and x hasn't been quantified.

Conjecture:

What happens if we don't know the Truth Value of a statement?

Consider the claim:

“If you repeatedly apply the following rule to any positive integer (divide by 2 if even, multiply by 3 and add 1 if odd), you will eventually reach 1”

What can you say about the Truth value of the above claim?

We do not know whether the claim is True or False, but we can certainly say that only one of those truth values applies. i.e. the claim cannot be both True as well as False . It can either be true or it can be false. In other words, this claim works for every positive integer or there is at least one positive integer for which is claim is untrue in which case this entire claim becomes False. This is famously known as “ Collatz Conjecture”

Can we then call this claim a mathematical statement?

Yes, in fact, this claim does satisfy the properties of a mathematical statement.

This means, even if we do not know whether a statement is True or False, it can still be a mathematical statement.

This leads to our next definition:

Conjecture: *In mathematics, a conjecture is a mathematical statement which appears to be true, but has not been formally proven.*

Practice:

Are the following mathematical statements? Why/why not?

1. $x \leq 5$ and $x > 3$ *variable proposition*
2. $\exists x \in \mathbb{R}. x \leq 5$ and $x > 3$ *Mathematical Statement*
3. Any even natural number greater than or equal to 4 can be written as the sum of two prime numbers. *Conjecture $\rightarrow$ M.S.*

Quantifiers:

Quantifiers can be divided into 2 categories:

Universal Quantifier: *As the name suggests, Universal Quantifier states that the claim holds for every element in a set. Typically, as we have seen before, the notation used for Universal Quantifier is $\forall$, read as “For All”.*

For example,

Quantified
 $\forall x \in \mathbb{R}. x^2 \geq 0$ *$\rightarrow$ claim*

For the above mathematical statement to be true, every element, x , in the set of Real numbers must satisfy the property that their square is greater than or equal to 0.

Existential Quantifiers: *Existential Quantifier states that the claim holds for at least one element in the set. The notation used for existential quantifier is $\exists$, read as “there exists”. For example:*

* We generally separate multiple Quantifiers using "and".

$$\exists x \in \mathbb{R}. x^2 \geq 4$$

For this mathematical statement to be True, we need to show that there is at least one element, x , in the set of real numbers which satisfies the property that their square is greater than or equal to 4. There may be other real numbers which do not satisfy the claim but as long as we have at least one real number that does, the above statement is True.

Let's try a few examples:

How would you rewrite the following using mathematical symbols?

1. *The square of a real number is always a real number.* $\forall x \in \mathbb{R}. x^2 \in \mathbb{R}$
2. *Every natural number is greater than or equal to 1* $\forall x \in \mathbb{N}. x \geq 1$
3. *Every integer exceeds another* $\forall x \in \mathbb{Z} \exists y \in \mathbb{Z}. x > y$

Commonly used number sets in Mathematics

$\mathbb{R}$: Set of real numbers: Any value on the number line

$\mathbb{N}$: Set of Natural Numbers: Whole numbers from 1 upwards

$\mathbb{Z}$: Integers: Whole numbers and negative whole numbers including 0

$\mathbb{Q}$: Rational Numbers: Numbers you can make by dividing one integer by another (except dividing by 0)

$\mathbb{I}$: Imaginary numbers: Numbers that when squared give negative results

$$\forall x \in \mathbb{Z}. \exists y \in \mathbb{Z}. x > y$$

$$\exists y \in \mathbb{Z}. \forall x \in \mathbb{Z}. x > y$$

What is the difference?